

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**

Lecture No.– 09



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Topics to be Covered

Topic



- 1:- computer Hardware ✓
- 2:- Types of Computer Hardware ✓
- 3:-INPUT , OUTPUT , STORAGE DEVICES ✓
- 4:- Storage Device – Secondary Memory ✓
- 5:- Storage Device – Primary Memory ✓

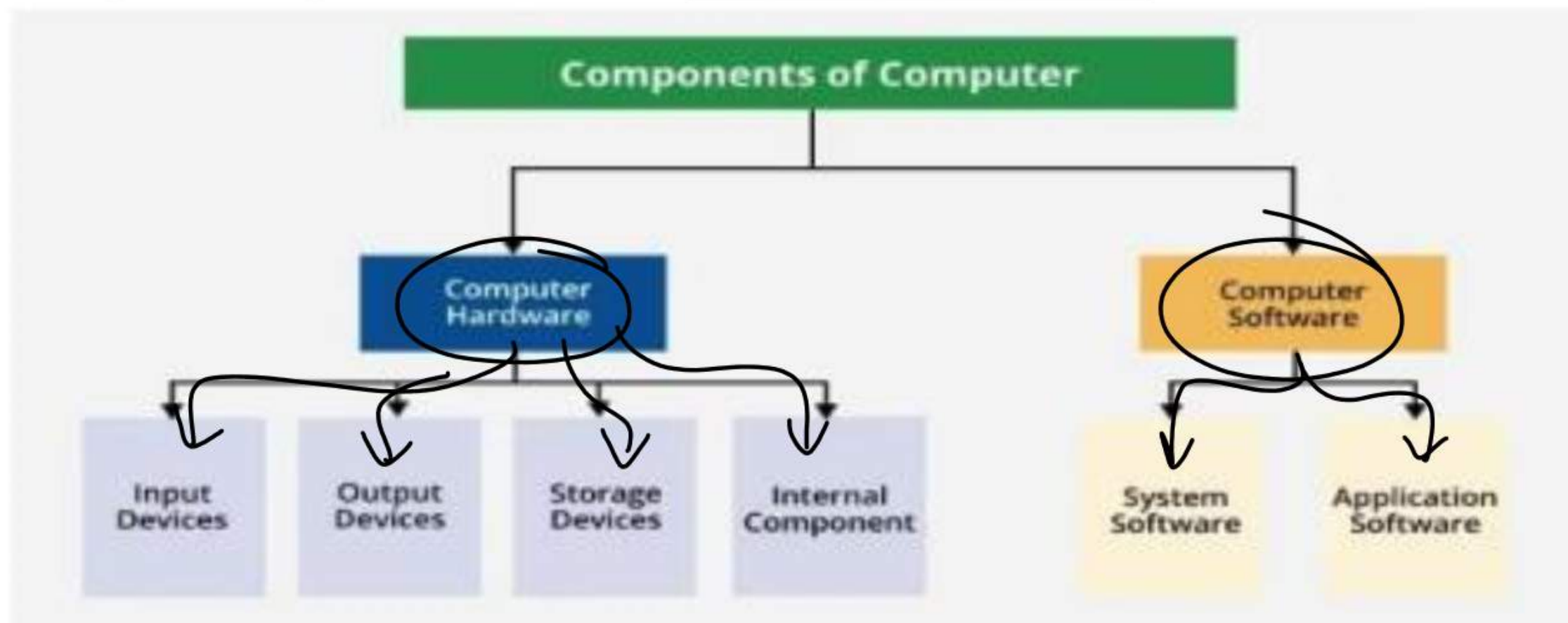


Topic: Computer Hardware

Computer hardware refers to the physical components of a computer that you can see and touch. These components work together to process input and deliver output based on user instructions. In this article, we'll explore the different types of computer hardware, their functions, and how they interact to make your computer work.

What is a Computer Hardware

Computer hardware is a physical device of computers that we can see and touch. E.g. Monitor, Central Processing Unit, Mouse, Joystick, etc. Using these devices, we can control computer operations like input and output.





Keyboard: The most widely used input device, featuring 104 keys, including alphabetic, numeric, and function keys. Modern keyboards connect via Bluetooth, replacing traditional wired connections.

||·|||N Asc Value

Mouse: A pointing device that controls the cursor on the screen. It features left, right, and middle buttons for selection and interaction. The sensor inside the mouse detects its movement speed, adjusting the cursor accordingly.

Scanner: Scans documents, images, and other media, converting them into digital formats for editing or processing, similar to a Xerox machine.

Microphone: Converts sound into electrical signals. It captures voice input for speech recognition and voice commands on the computer.

||| |||

Optical Character Reader (OCR): Scans printed or handwritten text, converting it into digital data by detecting reflected light from the characters, similar to a scanner.

Bar Code Reader: Reads bar codes and converts them into digital data for processing. The bar code consists of light and dark lines that encode information.



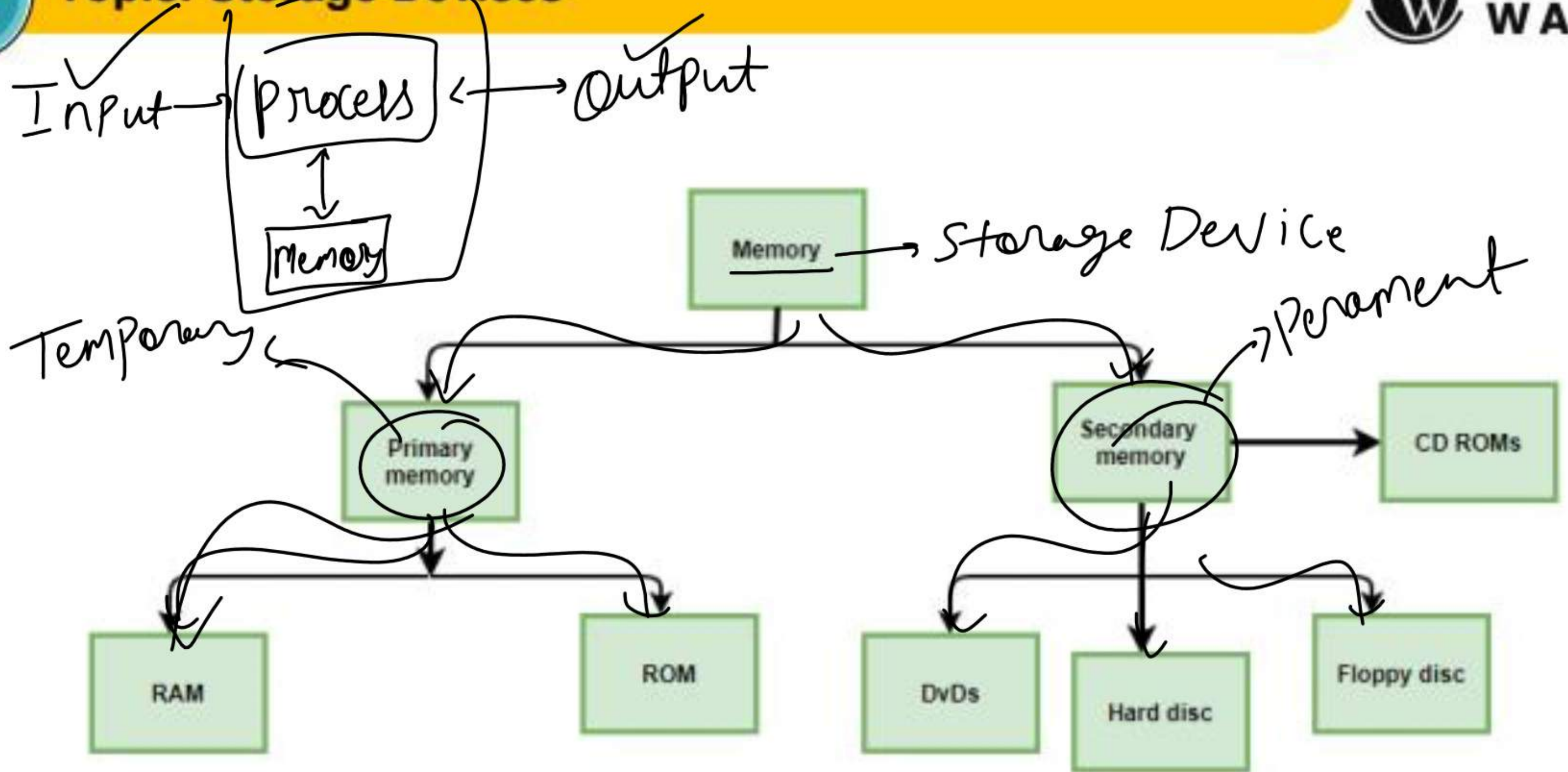
- **Monitor:** The main output device. It is also called VDU (visual display unit) and it looks like a TV screen. The Monitor displays the information from the computer. It is used to display text, video, images, etc.
- **Printer:** A printer is an output device that transfers data from the computer in a printed format by using text or images on paper. There are both coloured and black & white printers. Further, there are also different types of printers, like Laser Printer, Dot-matrix printers, and Inkjet printers.
- **Plotter:** It is similar to a printer but plotters are large. A plotter is used to generate large drawings, architectural blueprints, etc. on paper and these are high-quality images and drawings and large.
- **Speakers:** It is a very common output device and it gives sound as an output. The speaker is generally used to play music or anything having sound.



Topic: Storage Devices



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Topic: Storage Devices – Secondary Memory

Some devices are used for storage purposes and are known as secondary storage devices. Some of them are discussed below:

1. CD (Compact disc): A CD is circular and made up of thin platted glass and plastic polycarbonate material. It has a storage capacity of 600 MB to 700 MB of data. It has a standard size of 12 cm with a hole in the centre of about 1.5 cm and 1.2 mm in thickness. There are 3 types of CDs, which are:

- **CD-ROM (CD - Read Only Memory):** Contents of this type of CD cannot be erased by the user. Only the publisher is allowed to access the data imprinted on this CD. CD-ROM is used for commercial purposes like for a music album or any application package by a software company.
- **CD-R (CD-Recordable):** In this, content or data can be stored once. After that, they can be read many times but the data or content cannot be rewritten or erased. (Kind of one-time use)
- **CD-RW (CD-Rewritable):** As the name suggests, this type of CD is used to rewrite the content or erase previous content and again write new content many times.



2. **DVD (Digital Video/Versatile Disc):** A DVD is the same as a CD but with some more features. A DVD comes in single and dual-layer formats. It has much greater storage capacity in comparison to CD. The storage capacity of a DVD with a one-sided single layer is - 4.7 GB, one-sided double layer - 8.5 GB, double-sided single layer - 9.4 GB, and double-sided double layer - 17 GB. There are also some types of DVDs, which are :

- **DVD-ROM:** *Already written* In this type, the contents of the DVD cannot be written on or erased by the user. DVD ROM is used for applications and databases for distributing them in large amounts.



- **DVD-RAM:** DVD RAM is accessed like a hard disk. It provides high data security and storage capacity. This is a kind of rewritable disc and it allows up to 1,00,000 rewrites.

Power ↑



Topic: Storage Devices - Secondary Memory

3. **Hard Disk** An hard disk is a non-volatile storage device that uses its read/write heads to store digital data on the magnetic surface of a rigid plate. It is generally 3.5 inches in size for desktops and 2.5 inches in size for laptops. A hard disk can be classified further into 3 types, which are:

MB $\rightarrow 10^6$ Bytes
 $\rightarrow 10^9$ Bytes
 $\rightarrow 10^{12}$ Bytes

- Internal Hard Disk: It has a common storage capacity stated as GB or TB. A system case or cabinet is the place where it is located. It can perform faster operations and its storage is fixed. It is mainly used to store large data files and programs.

- Hard Disk Packs are used by organizations such as banks and government sector organizations to store large amounts of data. They have a storage capacity of PB (peta bytes).

10^{15} Bytes

1 Bytes = 8 bits

1 bit Storage $\Rightarrow$ Register

volatile

Power off
Data Can
Erased



ROM, or Read-Only Memory, is a type of computer memory that stores data permanently and cannot be easily modified. It's used to hold firmware, which are the instructions that tell a computer how to perform basic functions like starting up and initializing hardware. ROM is non-volatile, meaning it retains its data even when the power is turned off.

Storing Firmware:

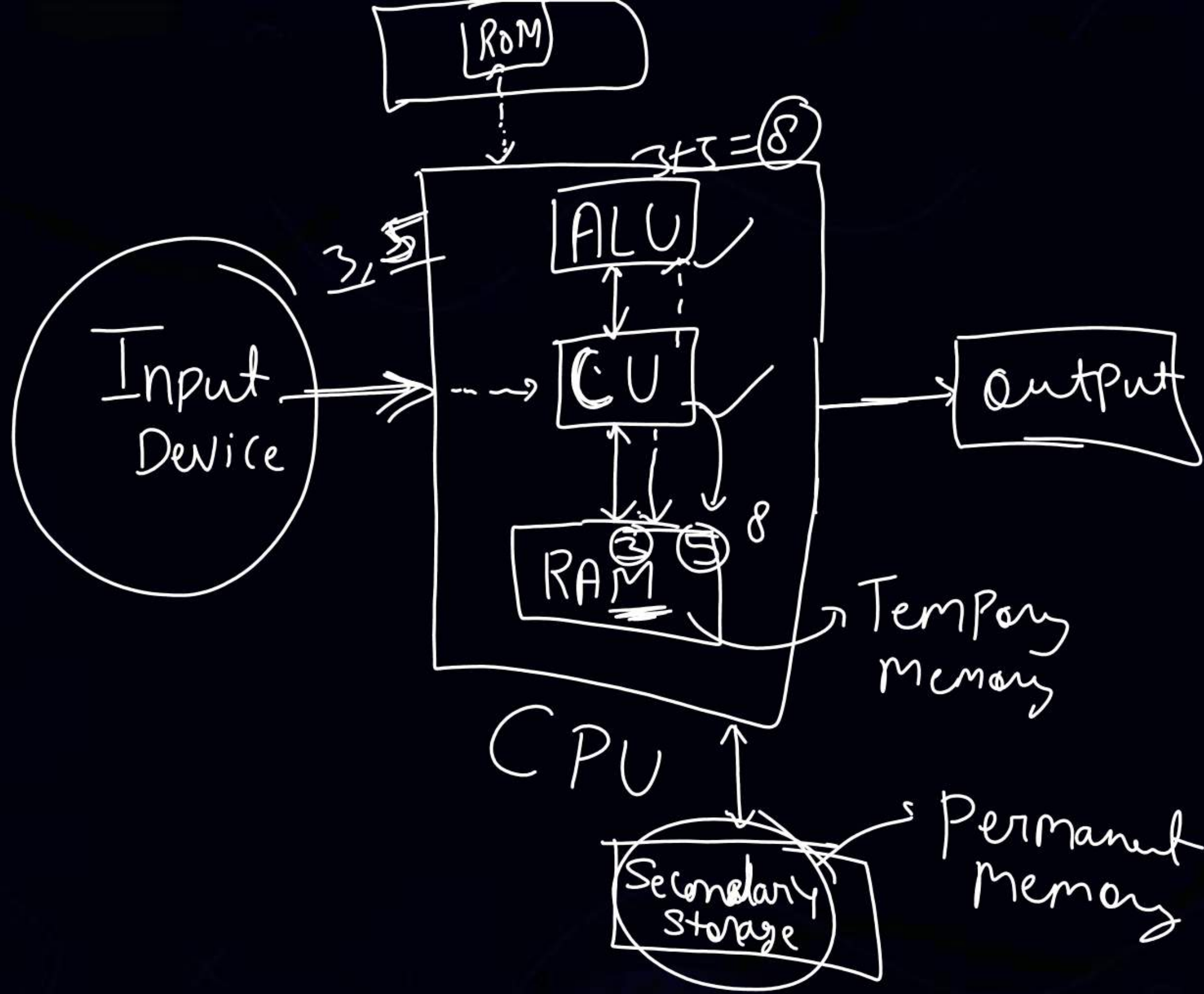
ROM is primarily used to store firmware, the low-level software that's essential for a computer to boot up and operate. This includes the BIOS (Basic Input/Output System) on PCs, which handles the initial start up sequence.

Non-Volatile Memory:

Unlike RAM (Random Access Memory), ROM is non-volatile, meaning the data stored in it is not lost when the power is turned off. This is crucial for storing essential boot instructions and other firmware that needs to be available every time the computer starts.

→ Power → Manufacturer

Non volatile





Evolution of ROM Technology

The development of ROM has seen key advancements over the years:

Year	Type	Key Advancement	Use Cases
1956	Mask ROM (MROM)	Hardwired during manufacturing	Early calculators, embedded systems
1956	PROM	One-time programmable by users	Custom firmware
1971	EPROM	Erasable with UV light, reprogrammable	Legacy computer BIOS
1983	EEPROM	Electrically erasable, reusable	Microcontrollers, car key fobs
1984	Flash Memory	Block-level erasure, high speed	USB drives, SSDs, smartphones

Desktop/Laptop
mobile
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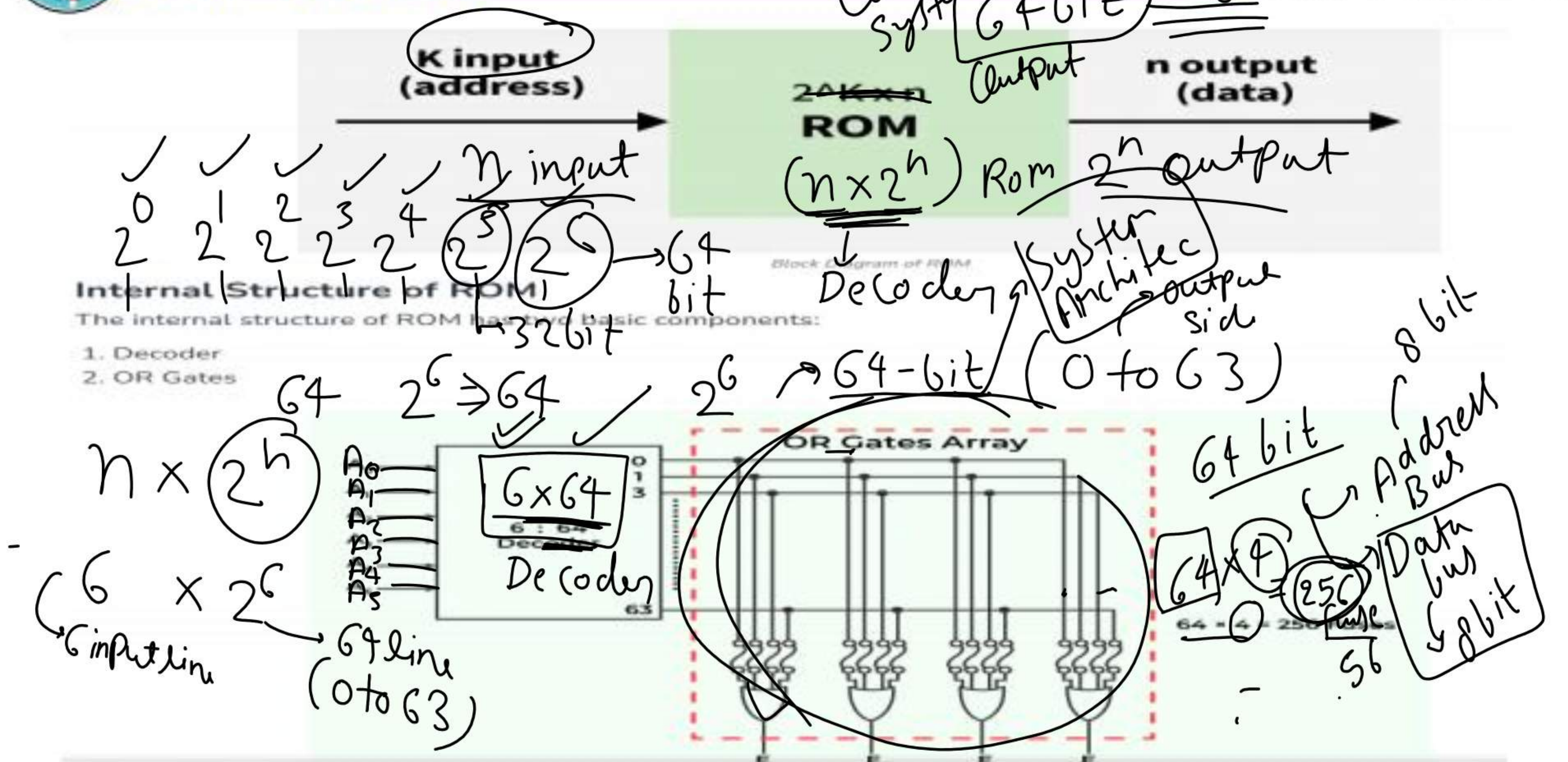


Types of Read-Only Memory (ROM)

ROM Type	Erasure Method	Reprogrammable	Use Cases/Examples
Mask ROM (MROM)	Hardwired during manufacturing	No	Early embedded systems, firmware
PROM	One-time programming	No	Custom firmware for specific applications
EPROM	UV light	Yes (with UV)	Firmware updates, legacy computer systems
EEPROM	Electrical signals	Yes	Microcontrollers, BIOS, small firmware updates
Flash Memory	Block-level electrical erasure	Yes	USB drives, SSDs, memory cards, smartphones
PLD-ROM	Configurable logic	Yes	FPGA, CPLD, custom hardware logic



1. **MROM (Masked read-only memory):** We know that ROM is as old as semiconductor technology. MROM was the very first ROM that consisted of a grid of word lines and bit lines joined together by transistor switches. This type of ROM data is physically encoded in the circuit and only be programmed during fabrication. It was not so expensive.
2. **PROM (Programmable read-only memory):** PROM is a form of digital memory. In this type of ROM, each bit is locked by a fuse or anti-fuse. The data stored in it are permanently stored and can not be changed or erasable. It is used in low-level programs such as firmware or microcode.
3. **EPROM (Erasable programmable read-only memory):** EPROM also called EROM, is a type of PROM but it can be reprogrammed. The data stored in EPROM can be erased and reprogrammed again by ultraviolet light. Reprogrammed it is limited. Before the era of EEPROM and flash memory, EPROM was used in microcontrollers.
4. **EEPROM (Electrically erasable programmable read-only memory):** As its name refers, it can be programmed and erased electrically. The data and program of this ROM can be erased and programmed about ten thousand times. The duration of erasing and programming of the EEPROM is about 4ms to 10ms. It is used in microcontrollers and remote keyless systems.





Topic: Advantage and Disadvantage of ROM

Advantages of ROM

- **Non-Volatile** – Retains data without power.
- **Security** – Prevents unauthorized changes.
- **Reliable** – Data remains intact over time.
- **Cost-Effective** – Cheap for large-scale production.
- **Fast Access** – Quick retrieval of stored data.

Disadvantages of ROM

- **Limited Modifiability** – Data can't be easily changed.
- **Low Storage Capacity** – Not suitable for large data storage.
- **Slow Write Speeds** – Writing data is slow.
- **Physical Wear** – Can wear out after many write cycles (EEPROM/EPROM).
- **High Initial Cost** – Expensive to manufacture in small quantities (Mask ROM).



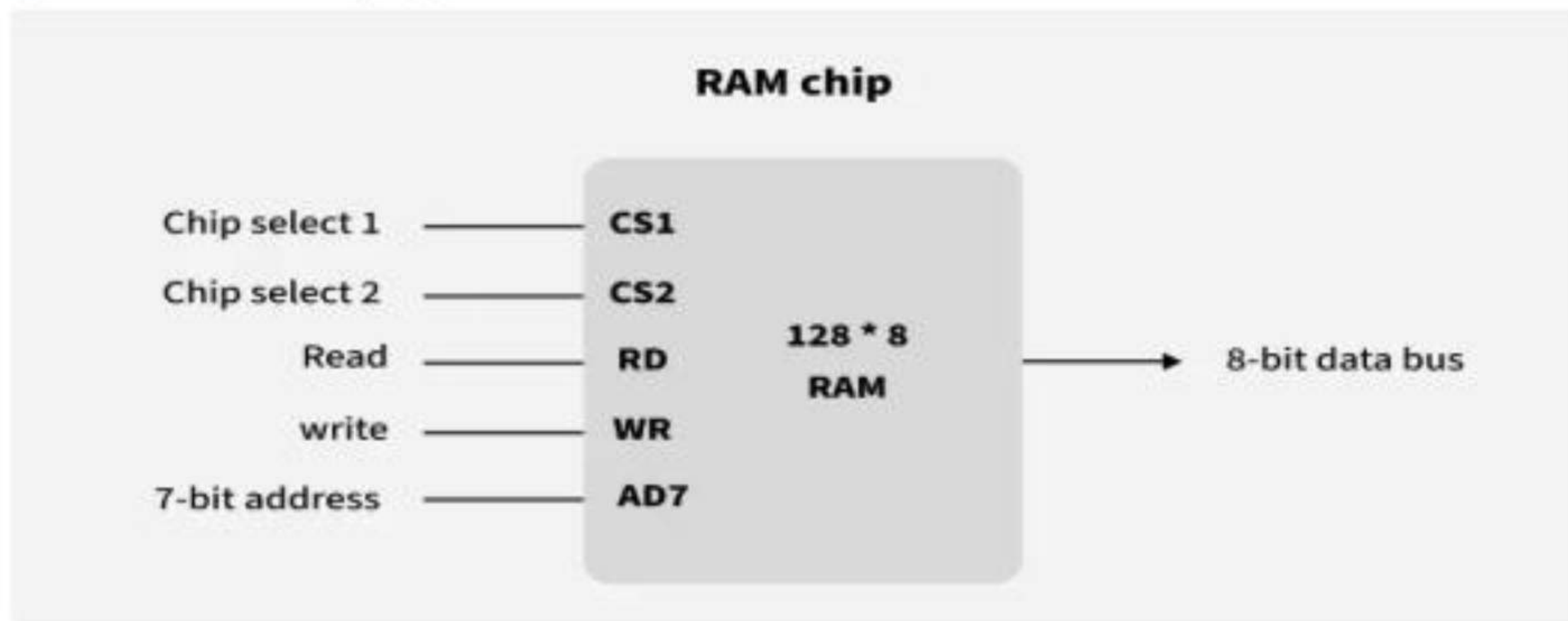
Topic: Primary Memory- RAM

What is RAM?

Random Access Memory, is a type of computer memory that allows data to be read and written randomly, meaning that the computer can access any location in the memory directly rather than having to read the data in a specific order. This makes RAM an essential component of a computer system, as it enables the CPU to access data quickly and efficiently.

RAM is volatile in nature, which means if the power goes off, the stored information is lost. RAM is used to store the data that is currently processed by the CPU. Most of the programs and data that are modifiable are stored in RAM.

The block diagram of the RAM chip is given below:

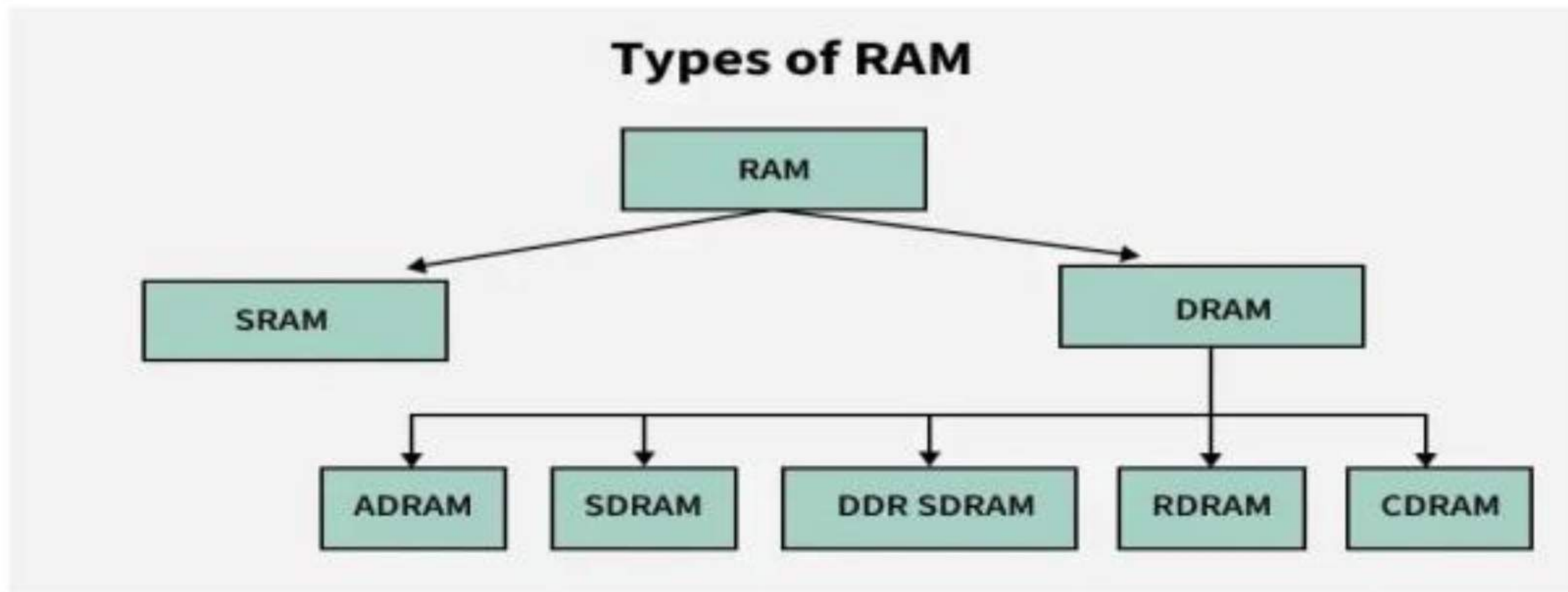




Types of RAM?

Mainly RAM have 2 types

- SRAM (Static RAM)
- DRAM (Dynamic RAM)





Topic: Primary Memory- RAM

Types of RAM

RAM is further divided into two types, SRAM - Static Random Access Memory and DRAM- Dynamic Random Access Memory. Let's learn about both of these types in more detail.

1. SRAM (Static Random Access memory)

SRAM is used for Cache memory, it can hold the data as long as the power availability is there. It is refreshed simultaneously to store the present information. It is made with CMOS technology. It contains 4 to 6 transistors and it also uses clocks. It does not require a periodic refresh cycle due to the presence of transistors. Although SRAM is faster, it requires more power and is more expensive. Since SRAM requires more power, more heat is lost here as well, another drawback of SRAM is that it can not store more bits per chip, for instance, for the same amount of memory stored in DRAM, SRAM would require one more chip.

Function of SRAM

The function of SRAM is that it provides a direct interface with the Central Processing Unit at higher speeds.

Characteristics of SRAM

- SRAM is used as the Cache memory inside the computer.
- SRAM is known to be the fastest among all memories.
- SRAM is costlier.
- SRAM has a lower density (number of memory cells per unit area).
- The power consumption of SRAM is less but when it is operated at higher frequencies, the power consumption of SRAM is compatible with DRAM.



2. DRAM (Dynamic Random Access memory)

DRAM is used for the Main memory, it has a different construction than SRAM, it uses one transistor and one capacitor (also known as a conductor), which is needed to get recharged in milliseconds due to the presence of the capacitor. Dynamic RAM was the first sold memory integrated circuit. DRAM is the second most compact technology in production (the First is Flash Memory). DRAM has one transistor and one capacitor in 1 memory bit. Although DRAM is slower, it can store more bits per chip, for instance, for the same amount of memory stored in SRAM, DRAM requires one less chip. DRAM requires less power and hence, less heat is produced.

Function of DRAM

The function of DRAM is that it is used for programming code by a computer processor to function. It is used in our PCs (Personal Computers).

Characteristics of DRAM

- DRAM is used as the Main Memory inside the computer.
- DRAM is known to be a fast memory but not as fast as SRAM.
- DRAM is cheaper as compared to SRAM.
- DRAM has a higher density (number of memory cells per unit area)
- The power consumption by DRAM is more



Difference Between SRAM and DRAM

Feature	SRAM (Static RAM)	DRAM (Dynamic RAM)
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Power Consumption	Requires more power	Requires less power
Cost	More expensive	Less expensive
Speed	Faster due to no need for refreshing	Slower because it needs to be refreshed
Usage	Used in cache memory for quick access	Used in main memory for large data storage



ROM vs RAM

ROM	RAM
Non-volatile memory used for permanent storage	Volatile memory used for temporary storage
Generally slower than RAM	High-speed access
Primarily read-only	Read and write operations
ROM can hold more than just megabytes	RAM can be stored in gigabytes
Data accessible is less difficult but more restrictive	Data accessibility is easy
Cheaper than RAM	High cost as compared to ROM
Used for the permanent storage of data	Used for temporary storage of data



Topic: Primary Memory- RAM

Advantages of RAM

- **Speed:** RAM is faster than other types of storage like ROM, hard drives or SSDs, allowing for quick access to data and smooth performance of applications.
- **Multitasking:** More RAM allows a computer to handle multiple applications simultaneously without slowing down.
- **Flexibility:** RAM can be easily upgraded, enhancing a computer's performance and extending its usability.
- **Volatile Storage:** RAM automatically clears its data when the computer is turned off, reducing the risk of unwanted data accumulation.

Disadvantages of RAM

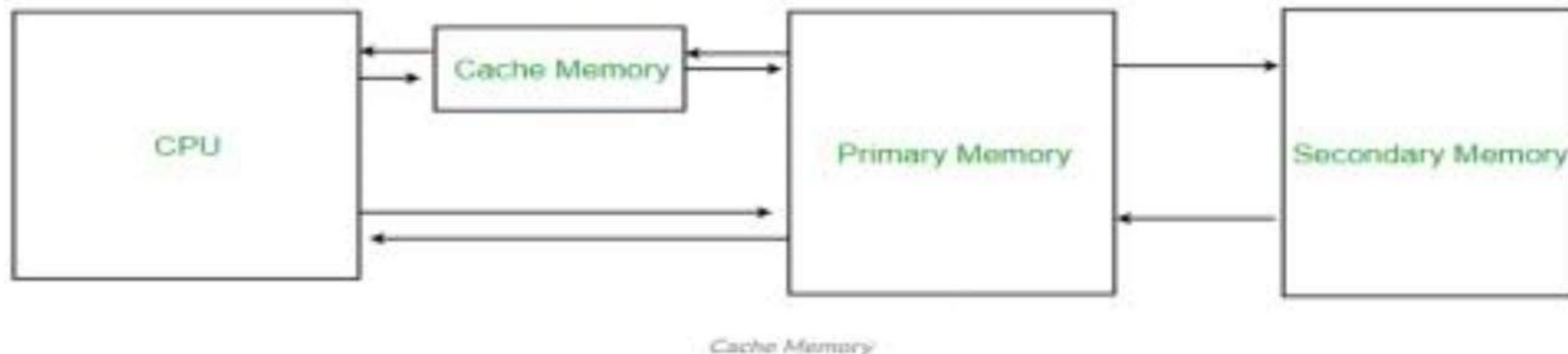
- **Volatility:** Data stored in RAM is lost when the computer is turned off, which means important data must be saved to permanent storage.
- **Cost:** RAM can be more expensive per gigabyte compared to other storage options like hard drives or SSDs.
- **Limited Storage:** RAM has a limited capacity, so it cannot store large amounts of data permanently.
- **Power Consumption:** RAM requires continuous power to retain data, contributing to the overall power consumption of the device.
- **Physical Space:** Increasing RAM requires physical space in the computer, which might be limited to smaller devices like laptops and tablets.



Topic: Cache Memory – Static RAM

Characteristics of Cache Memory

- Extremely fast memory type that acts as a buffer between [RAM](#) and the CPU.
- Holds frequently requested data and instructions, ensuring that they are immediately available to the CPU when needed.
- Costlier than main memory or disk memory but more economical than CPU registers.
- Used to speed up processing and synchronize with the high-speed CPU.



Levels of Memory

- **Level 1 or Register:** It is a type of memory in which data is stored and accepted that are immediately stored in the CPU. The most commonly used register is Accumulator, [Program counter](#), Address Register, etc.
- **Level 2 or Cache memory:** It is the fastest memory that has faster access time where data is temporarily stored for faster access.
- **Level 3 or Main Memory:** It is the memory on which the computer works currently. It is small in size and once power is off data no longer stays in this memory.
- **Level 4 or Secondary Memory:** It is external memory that is not as fast as the main memory but data stays permanently in this memory.



Topic: Cache Memory – Static RAM

Advantages

- Cache Memory is faster in comparison to main memory and secondary memory.
- Programs stored by Cache Memory can be executed in less time.
- The data access time of Cache Memory is less than that of the main memory.
- Cache Memory stores data and instructions that are regularly used by the CPU, therefore it increases the performance of the CPU.

Disadvantages

- Cache Memory is costlier than primary memory and secondary memory.
- Data is stored on a temporary basis in Cache Memory.
- Whenever the system is turned off, data and instructions stored in cache memory get destroyed.
- The high cost of cache memory increases the price of the Computer System.

THANK YOU